Hall Ticket Number:

MR-21

Code: B415CS4S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

Java Programming

(Common to CSE and IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Explain the purpose of the "this" keyword in Java.	2	2	1	1
1. b)	Write a simple Java class with a constructor and a method that displays student details.	2	3	1	1,2
1. c)	Explain the difference between abstract class and interface.	2	2	2	1
1. d)	What is an exception? How does it differ from an error?	2	2	2	1
1. e)	What is process-based multitasking?	2	1	3	1
1. f)	Explain various string handling classes available in Java.	2	1	3	1
1. g)	List any four main interfaces of the Collections Framework.	2	1	4	1
1. h)	Write a Java snippet to iterate through an ArrayList using an Iterator.	2	3	4	1,2
1. i)	Explain the purpose of the paint() method in applets.	2	2	5	1
1. j)	List any two components of the java.awt package.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	What are Java buzzwords? Explain any five with suitable examples.	5	1	1	1
2. b)	Discuss the role of constructors and method overloading in Java with examples.	5	2	1	1
	OR				
2. c)	Discuss various control statements in Java with suitable examples.	5	2	1	1
2. d)	Differentiate between compile-time and runtime polymorphism in Java.	5	4	1	1,2
3. a)	Explain the process of defining, creating, and accessing packages in Java.	5	1	2	1
3. b)	Write a Java program to handle multiple exceptions using multiple catch clauses.	5	3	2	1,2
	OR				
3. c)	Discuss the importance of access protection and package visibility with examples.	5	2	2	1
3. d)	Compare and contrast throw and throws keywords in Java with examples.	5	4	2	1,2
4. a)	Explain thread priorities and their role in Java thread scheduling.	5	2	3	1
4. b)	Describe different methods of the String class with examples.	5	1	3	1
	OR				
4. c)	Develop a Java program demonstrating inter-thread communication using wait(), notify(), and notifyAll().	5	3	3	1,2,3
4. d)	Explain the use and methods of the File class in Java.	5	2	3	1
5. a)	Discuss the importance of the Map interface and explain HashMap and TreeMap.	5	2	4	1
5. b)	Describe how the StringTokenizer class is used for string parsing.	5	2	4	1
	OR				

P.T.O.

5. c)	Write a Java program to sort a list of names using a Comparator.	5	3	4	1,2
5. d)	Write a Java program that uses Scanner to read data and store it into a LinkedList.	5	3	4	1,2
6. a)	Demonstrate the delegation event model in Java with an example.	5	3	5	1,2
6. b)	Write a Swing applet to draw a rectangle and a circle on the screen.	5	3	5	1
	OR				
6. c)	How can parameters be passed to an applet using the <param> tag? Give a code example.	5	3	5	1,2
6. d)	Explain the working of FlowLayout, BorderLayout, and GridLayout with examples.	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-21

Code: B415CB4S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

Object Oriented Programming Using Java

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List any four principles of Object-Oriented Programming.	2	1	1	1
1. b)	Explain the difference between method overloading and overriding with an example.	2	2	1	1
1. c)	Define a package in Java. What are the steps to import a user-defined package?	2	2	2	1
1. d)	Illustrate the use of throws keyword in exception handling.	2	3	2	2
1. e)	What is the purpose of thread synchronization?	2	2	3	1
1. f)	Differentiate between byte stream and character stream.	2	2	3	2
1. g)	Define Iterator. How is it used in a Collection?	2	2	4	1
1. h)	Write the difference between HashMap and TreeMap.	2	2	4	1
1. i)	What is event delegation? Give one example.	2	2	5	2
1. j)	Define LayoutManager and mention any two of its types.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the structure of a Java program and the role of JVM and JDK.	5	2	1	1
2. b)	Write a Java program to demonstrate the use of "this" and "super" keywords.	5	2	1	1,2
	OR				
2. c)	Discuss access modifiers in Java with examples.	5	2	1	1
2. d)	Write a Java program to demonstrate single and multilevel inheritance.	5	2	1	1,2
3. a)	Explain how interfaces support multiple inheritance in Java.	5	2	2	1
3. b)	Develop a Java program implementing multiple interfaces for calculating area and volume.	5	3	2	1,2
	OR				
3. c)	Explain different types of exceptions in Java.	5	2	2	1
3. d)	Develop a Java program using try-catch-finally to handle ArrayIndexOutOfBoundsException.	5	3	2	1,2,3
4. a)	Explain the thread life cycle and how priorities affect scheduling.	5	2	3	1
4. b)	Write a Java program to create two threads and run them concurrently.	5	3	3	1,2
	OR				
4. c)	Explain file handling using FileReader and BufferedReader classes.	5	2	3	1
4. d)	Develop a Java program to count the number of words in a text file.	5	3	3	1,2
5. a)	Categorise the major interfaces in the Java Collections framework? Explain any three.	5	3	4	1
5. b)	Implement a Java program to demonstrate sorting using Collections.sort().	5	3	4	1,2

P.T.O.

	OR				
5. c)	Explain the role of HashSet, LinkedHashSet, and TreeSet with examples.	5	2	4	1
5. d)	Develop a Java program that counts unique words from a file using a HashSet.	5	4	4	1,2,3
6. a)	Discuss the architecture of Swing and its advantages over AWT.	5	2	5	1
6. b)	Develop a Java Swing program to display a calculator GUI.	5	4	5	1,2
	OR				
6. c)	Explain different layout managers used in GUI design.	5	2	5	1
6. d)	Develop a Java program to demonstrate mouse and keyboard event handling using adapter classes.	5	4	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome